

Infinite Groups 1994

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Proceedings of the International Conference,
held in Ravello, Italy, May 23–27, 1994

Editors

Francesco de Giovanni
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Preface

We present with pleasure this volume containing most of the talks delivered at the Ravello Conference held in May 1994. The vast scope and diversity of the contributions reflect the wide range of current investigations and developments in the theory of Infinite Groups. It also confirms our conviction that an International Conference on this topic was timely and that our joint decision to organise one was well worth while. Although the decision was ours, we acknowledge that the labour and burden of dealing with all the necessary arrangements was borne in most part by others in the Department of Mathematics in Napoli. For their heroic efforts we wish to thank most sincerely Rara Celentani, Giovanni Cutolo, Ulderico Dardano, Silvana Rinauro and Gianni Vincenzi.

It was extremely gratifying that the conference attracted such an impressive world-wide list of participants. The international renown and proven ability of our main speakers combined with the particular charm of the Ravello venue, were important factors. But also the range of interesting topics covered by contributors in the parallel afternoon sessions sustained the stimulating momentum and resulted in many late-night heated discussions in the Piazza. We thank them all for making the venture such a memorable success.

With particular appreciation we wish to gratefully acknowledge the financial support generously provided by the "Consiglio Nazionale delle Ricerche", the "Dipartimento di Matematica e Applicazioni dell'Università di Napoli Federico II", the "Ente Provinciale per il Turismo di Salerno", the "International Science Foundation", the "Istituto Italiano per gli Studi Filosofici", the "Ministero dell'Università e della Ricerca Scientifica e Tecnologica" and the "Università di Napoli Federico II".

To the civic authorities in Ravello we extend our appreciation for their hospitable welcome and courteous attention to the needs of so many foreign visitors.

We are confident that the publications contained in these collected proceedings will prove an important and significant contribution to the development of the study of infinite groups. Many of the deep questions raised will chart the way forward and provide the impetus for further years of mathematical endeavour. To Manfred Karbe and the staff of Walter de Gruyter & Co. we record our deepest appreciation, not only for their support, encouragement and cooperation while preparing this volume, but also for the provision of the book exhibition which provided a special focal point of interest during the conference.

In October we were deeply saddened by the sudden demise of one of our main speakers Professor Brian Hartley. His loss to the mathematical community is a great one. Few could have imagined that the relaxed, unassuming, good humoured personality who gave so much of his time and energy promoting the general conviviality of the occasion, would be so shortly taken from us. His ingenious techniques and mathematical insight provide a lasting source of inspiration for others to follow.

Finally we record our enormous debt of gratitude to Dr. Giovanni Cutolo who spent many hours checking and editing written contributions. We include of course the gallant band of referees who diligently assessed the papers and thank them for their professional evaluations and the promptness of their replies.

Every effort has been made to ensure that these proceedings contain an accurate reproduction of the articles submitted, but we beg your indulgence should any omissions nevertheless occur.

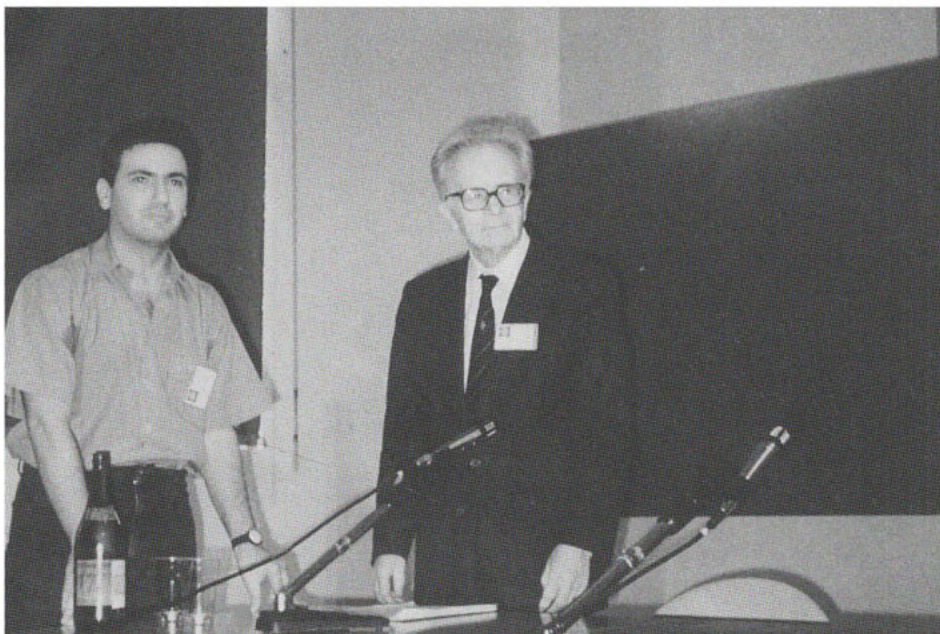
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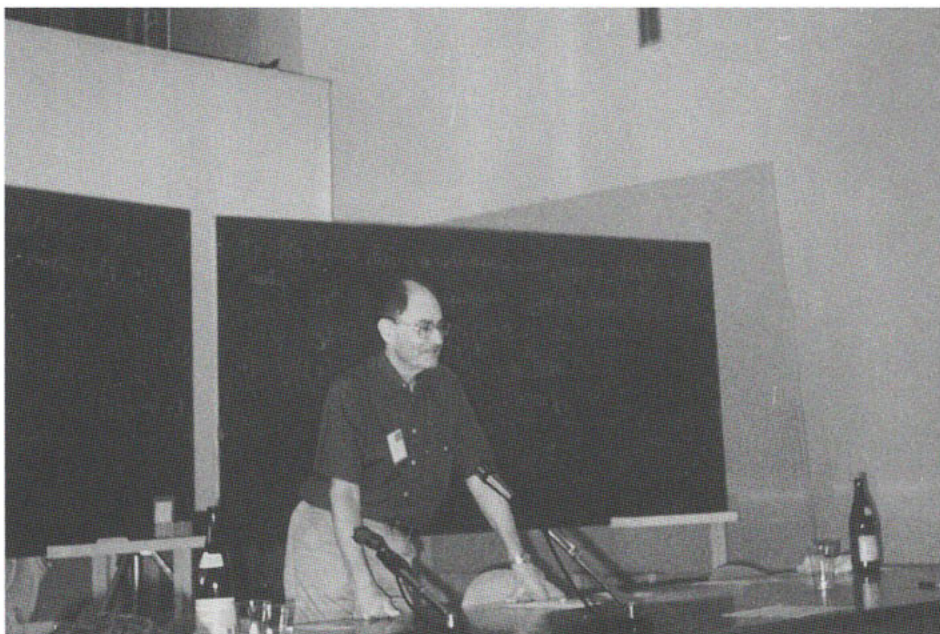
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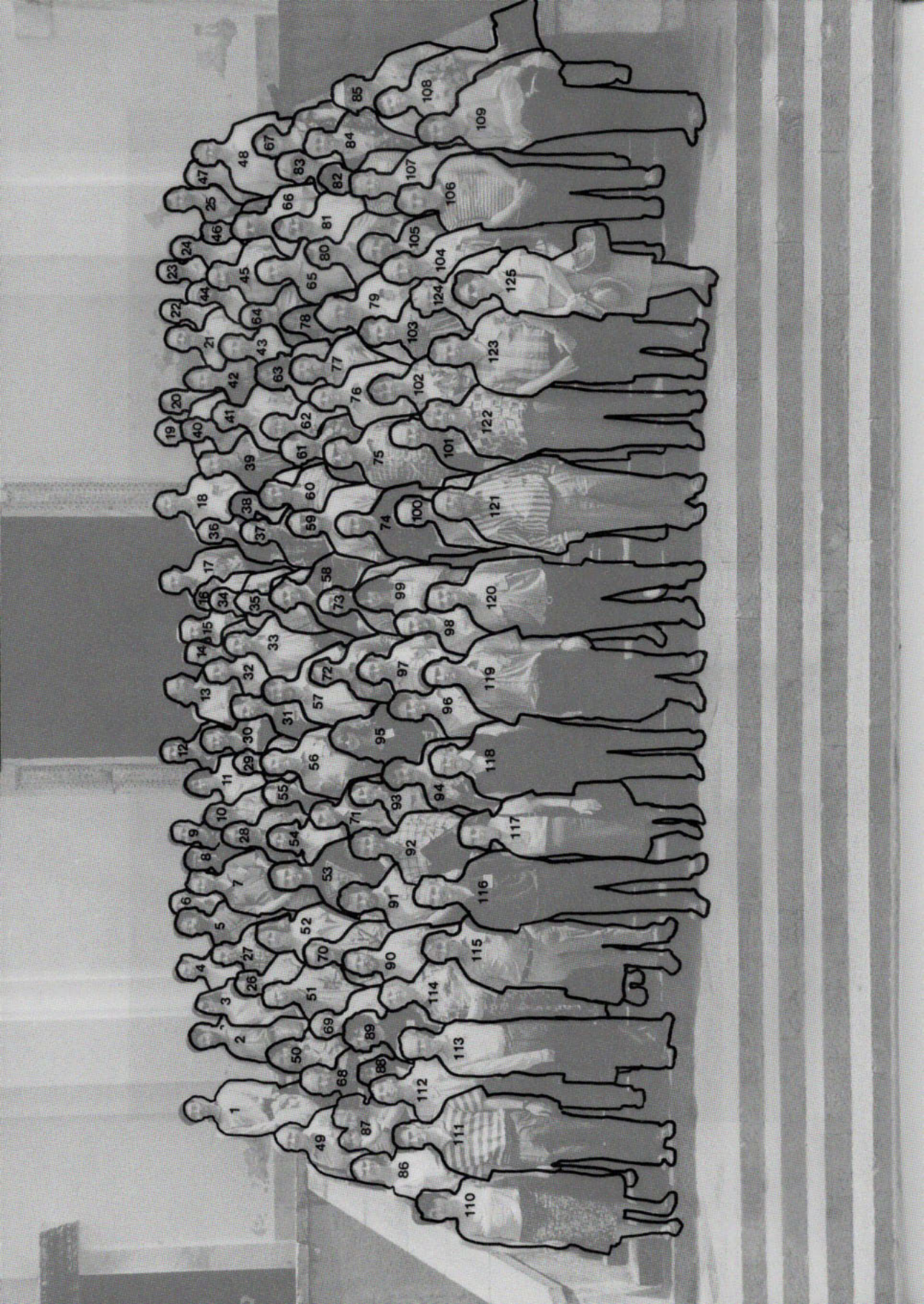
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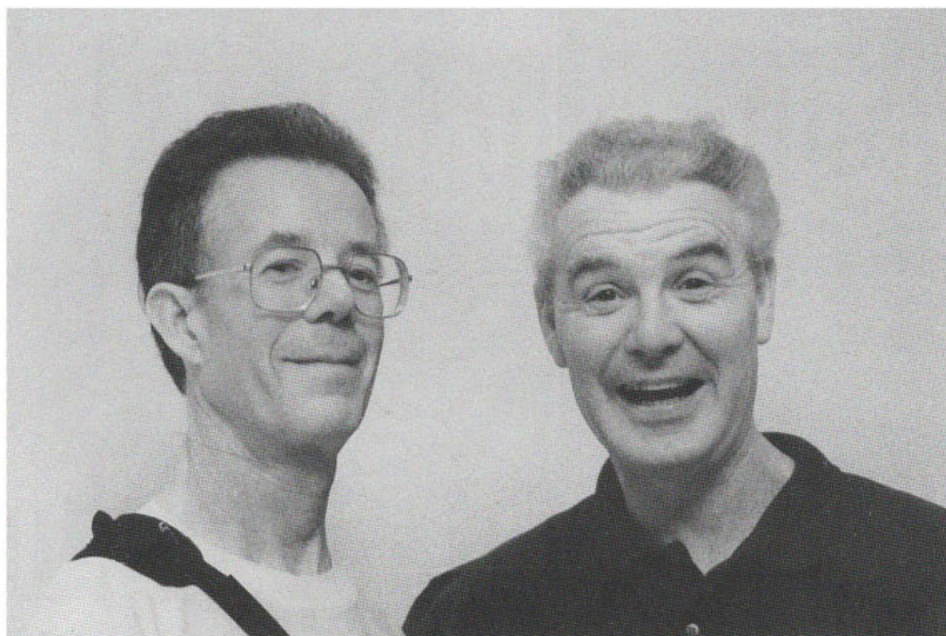


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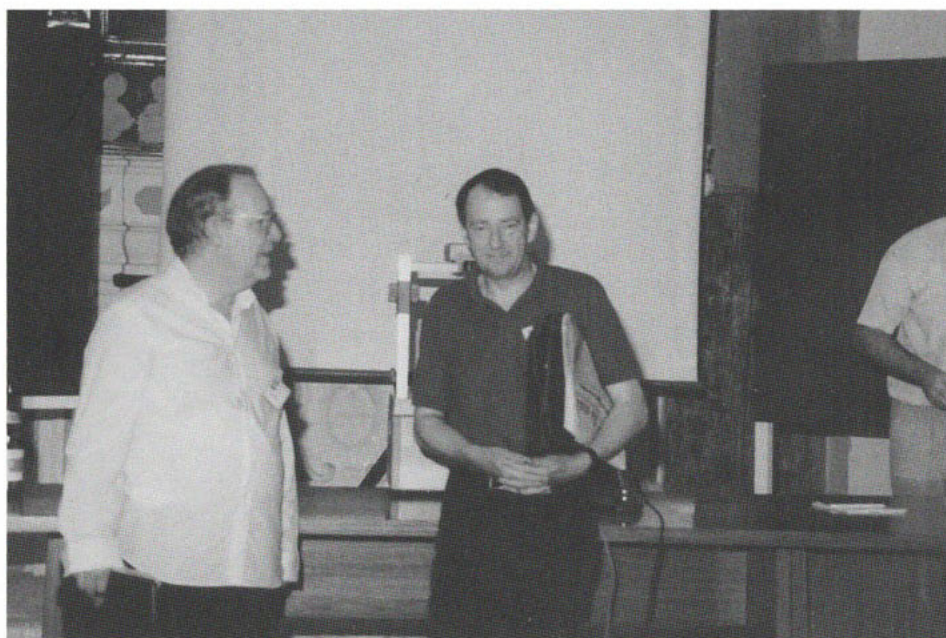


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Some Results and Problems about Factorized Groups

Bernhard Amberg

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1. Introduction

In the last years the structure of groups $G = AB$ which are the product of two of its subgroups A and B have received considerable attention. The most important results about mainly infinite such factorized groups are presented in the monograph [7]; see also [17] and the survey articles [2], [3], [35] and [42]. Among the central problems that were considered are the following:

(1) Let the group $G = AB$ be the product of two subgroups A and B and suppose that A and B satisfy some finiteness condition $\mathfrak{X}$. When does the factorized group G have the same finiteness condition $\mathfrak{X}$?

(2) Let the group $G = AB = AM = BM$ be the product of two subgroups A and B and a normal subgroup M of G and suppose that A , B and M satisfy some nilpotency condition $\mathfrak{N}$. When does the factorized group G have the same nilpotency condition $\mathfrak{N}$?

(3) Let the group $G = AB$ be the product of two subgroups A and B which satisfy some nilpotency condition $\mathfrak{N}$. What can be said about the structure of the factorized group G , when does it satisfy some solubility requirement? Which subgroups such as the Hirsch–Plotkin radical and the Fitting subgroup of $G = AB$ can be written as a product of a subgroup of A and a subgroup of B ?

The following article can be considered as an appendix to [7], to which we refer as a general reference. In particular we will discuss some new results with respect to the above problems which have been found since the appearance of this book and would have had an influence to our writing if we had known them at that time.

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2. Finiteness Conditions

2.1. General Products

Consider a group $G = AB$ which is the product of two subgroups A and B and suppose that $\mathfrak{X}$ is a group theoretical property which is inherited by subgroups, epimorphic images and extensions. If A and B are $\mathfrak{X}$ -groups, when does it follow that G likewise has the group theoretical property $\mathfrak{X}$? This problem is particularly interesting when $\mathfrak{X}$ is some finiteness condition.

The main finiteness conditions that were considered are the following:

A group G has *min* if it satisfies the minimum condition on subgroups. G has *max* if it satisfies the maximum condition on subgroups. G is a *minimax group* if it has a finite series whose factors satisfy the minimum or maximum condition for subgroups. G has *finite Prüfer rank* $r = r(G)$ if every finitely generated subgroup of G can be generated by r elements and r is the least such number. G has *finite abelian section rank* if every elementary abelian p -section of G is finite for every prime p . G has *finite torsion-free rank* if it has a series of finite length whose factors are periodic or infinite cyclic. The number of infinite cyclic factors in any such series is an invariant of G called its *torsion-free rank* $r_0(G)$. G is *periodic* if and only if it has torsion-free rank $r_0(G) = 0$.

The following theorem announced by Ivanov in [30] shows that without further assumptions none of the usual finiteness conditions is inherited by products of groups (see also [31], Theorem 34).

Theorem 2.1. *Let p and q be primes not less than 10^{78} , and let A and B be infinite groups all of whose non-trivial proper subgroups have order p and q , respectively. Then every countable group X can be embedded in a product $G = AB$ of A and B such that $A \cap B = 1$.*

Simple examples show that Theorem 2.1 has the following consequence.

Corollary 2.2. *None of the following finiteness conditions is inherited by products of groups: "min", "max", "minimax", "finite Prüfer rank", "finite abelian section rank", "periodic", "finite torsion-free rank".*

Using wreath products with respect to infinite sequences of permutation groups, Suchansky has also constructed residually finite groups $G = AB$ which are the product of two locally finite subgroups A and B , but which are not locally finite. Here G may be a p -group for some prime p , but also may have elements of infinite order; furthermore A may be a p -group and B a q -group for primes p and q with $p \neq q$ (see [47] and [48]).

There are many papers about the factorizations of finite simple groups. Factorizations of semisimple algebraic groups and of Lie groups were considered by Onishchik

in [38] and [39], Sections 5 and 14. In Belyaev [15] it is for instance shown that every countable locally finite simple group which is not a finitary linear group has many factorizations as a product of two residually finite groups.

2.2. Hyperabelian-by-Finite and Radical-by-Finite Groups

The situation is completely different if the group $G = AB$ satisfies some solubility requirement. For hyperabelian-by-finite groups the results obtained so far can be summarized in the following general theorem to which several authors have contributed; for details see [7], Chapters 3 and 4.

Theorem 2.3. *Let $\mathfrak{X}$ be one of the following finiteness conditions “min”, “max”, “minimax”, “finite Prüfer rank”, “finite abelian section rank”, “finite torsion-free rank” or “being a π -group for some set of primes π ”. If the hyperabelian-by-finite group $G = AB$ is the product of two $\mathfrak{X}$ -subgroups A and B , then G is an $\mathfrak{X}$ -group.*

The special case of Theorem 2.3 that every hyperabelian-by-finite product $G = AB$ of two subgroups A and B with finite torsion-free rank has likewise finite torsion-free rank was recently proved by Sysak in [53]. By [7], Theorem 4.1.8, the torsion-free rank satisfies the inequality $r_0(G) \leq r_0(A) + r_0(B) - r_0(A \cap B)$ in this case, but it is unknown whether equality holds here (see [7], section 4.1, in particular Question 8).

It is an interesting question for which other classes of generalized soluble groups a similar theorem holds. For instance, it would seem plausible that Theorem 2.3 extends to radical-by-finite groups. In fact, it follows from a result of Chernikov that a radical-by-finite product of two subgroups with minimum condition on subgroups likewise satisfies the minimum condition on subgroups (see [7], Theorem 3.2.12). Furthermore, Theorem 4 of Sysak [54] implies that a radical-by-finite product of two periodic groups is periodic. But it seems to be unknown at present whether similar results hold for the other finiteness conditions. In particular it is open whether a radical group $G = AB$ which is the product of two subgroups A and B with maximum condition likewise satisfies the maximum condition ([7], Section 4.4, Question 9). It is not difficult to see that we may reduce the consideration to the case that there exists a locally nilpotent normal subgroup M of G such that $G = AM = BM$.

2.3. Locally Soluble Groups

Since soluble products of polycyclic groups are polycyclic by Theorem 2.3, it is also clear that every locally soluble product of two groups with maximum condition on subgroups likewise satisfies the maximum condition on subgroups. A similar result holds for the minimum condition, since every product of two subgroups with minimum condition on subgroups satisfies the minimum condition for normal subgroups (see

[7], Lemma 1.2.6), and a locally soluble group with minimum condition for normal subgroups is known to be hyperabelian. On the other hand, Sysak has given examples to show that even a locally finite-soluble product of two subgroups with finite abelian section rank need not have finite abelian section rank. A general construction of such groups can be found in [54], Theorem 3.

In the following $\pi(X)$ denotes the set of primes p for which there is an element of order p in the group X .

Theorem 2.4 (Sysak [49]). *For every prime q there exists a locally finite-soluble group G of the form $G = A \ltimes V = AA^\alpha$, where V is a minimal normal subgroup of G which is an elementary abelian q -subgroup of G , A is a residually finite q' -subgroup of G and α is a locally inner automorphism of order q of G , which stabilizes the series $1 \triangleleft V \triangleleft G$; in addition, it may be assumed that A satisfies one of the following two conditions:*

- (a) $\pi(A) = \pi(q^n - 1)$ for any given natural number $n \geq 2$,
- (b) $q = 2$ and all Sylow subgroups of A are finite and abelian.

If, in particular, the group A in Theorem 2.4 satisfies condition (b), then the group G is the product of two subgroups with finite abelian section rank, but has infinite q -rank for the prime q . If the group A satisfies condition (a) and $q = 2$ and $2^n - 1$ is a Mersenne prime or $q = 3$ and $n = 2$, then G is the product of two p -subgroups, but itself has elements of two different prime orders.

A group satisfies the **weak minimum condition for subgroups** if every descending chain of subgroups S_i has only finitely many infinite indices $|S_{i+1} : S_i|$. The weak minimum condition for normal subgroups is defined similarly. It is well-known that a locally soluble minimax group is soluble and satisfies the weak minimum condition for subgroups. The following lemma is elementary.

Lemma 2.5 (Amberg [1]). *If the group $G = AB$ is the product of two subgroups A and B with weak minimum condition for subgroups, then G satisfies the weak minimum condition for normal subgroups.*

It follows from Theorem 2.1 that the group G in Lemma 2.5 the group G need not satisfy the weak minimum condition for subgroups. But for locally soluble groups the following can be said.

Theorem 2.6 (Amberg and Sysak [12]). *If the locally soluble group $G = AB$ is the product of two minimax subgroups A and B , then G is a soluble minimax group.*

Comments on the proof of Theorem 2.6. Recall that the standard polynomial of degree n is the polynomial

$$S_n(x_1, \dots, x_n) = \sum_{\pi \in \text{Sym}(n)} (\text{sgn } \pi) x_{\pi 1} \dots x_{\pi n}.$$

The ring R satisfies the standard polynomial of degree n if $S_n(r_1, \dots, r_n) = 0$ for all elements $r_1, r_2, \dots, r_n$ of R . It is easy to see that the property that a ring satisfies the standard polynomial for some degree n is inherited by subrings, factor rings and cartesian products. The Theorem of Amitsur and Levitzki says that the ring $M_n(R)$ of $n \times n$ -matrices with coefficients in the commutative ring R satisfies the standard polynomial of degree $2n$. The proof of Theorem 2.8 depends on the following proposition about the endomorphism ring of an abelian group of finite Prüfer rank which is of independent interest.

Proposition 2.7. *The endomorphism ring of an abelian group of finite Prüfer rank r satisfies the standard polynomial of degree $2r$.*

The proof of Theorem 2.6 reduces to the case of a hyperabelian group by the following theorem. Note that a group G is *residually of bounded finite Prüfer rank* if there exist normal subgroups N_i of G with $\bigcap N_i = 1$ and a positive integer k such that the Prüfer ranks r_i of the factor groups G/N_i satisfy $r_i \leq k$ for every i in the index set I . It is also used that a locally soluble subgroup of the multiplicative group of a ring with a polynomial identity is hyperabelian.

Theorem 2.8 (Amberg and Sysak[12]). *If the locally soluble group G is residually of bounded Prüfer rank, then G is hyperabelian.*

Using similar methods it can also be proved that every locally soluble product of two groups with finite Prüfer rank has likewise finite Prüfer rank. But it seems to be unknown at present whether a locally soluble product of two groups with finite torsion-free rank has likewise finite torsion-free rank.

Brookes and Smith have shown in [16] that if a group G is a product of two normal subgroups, each of which is locally soluble-of-finite-Prüfer-rank, then G is locally soluble if and only if it is locally of finite Prüfer rank. Since by a theorem of Robinson every finitely generated soluble group with finite Prüfer rank is a minimax group (see [41], Theorem 10.38), this implies that a locally soluble product of two normal locally minimax groups is a locally minimax group. It would be interesting to know whether this extends to arbitrary locally soluble products of locally minimax groups.

3. Triply Factorized Groups

3.1. Connections with Group Algebras

In the investigation of factorized groups one often encounters triply factorized groups of the form $G = AB = AM = BM$ where A , B and M are subgroups of the group G and M is normal in G . In many cases the group G in addition has the “intersection property” that $A \cap M = B \cap M = 1$, so that A and B are complements of M in G .

The following result establishes an interesting connection of triply factorized groups with group algebras; for a proof see [54], Theorem 1.

Theorem 3.1 (Sysak [51]). *Let G be a group and M a right kG -module for some commutative ring k . Then the following conditions are equivalent:*

- (1) *In the augmentation ideal $\Delta_k G$ there exists a right ideal $\mathfrak{p}$ of kG such that $\Delta_k G = G - 1 + \mathfrak{p}$, and the kG -module M is isomorphic with the kG -module $\Delta_k G/\mathfrak{p}$.*
- (2) *In the semi-direct product $G \ltimes M$ there exists a subgroup H such that $G \ltimes M = H \ltimes M = GH$.*

Moreover, for every subgroup H of G the right ideal $\mathfrak{p}$ can be chosen such that $G \cap H = G \cap (1 + \mathfrak{p})$.

In this situation also the following result of [13] can be proved.

Proposition 3.2. *Let G be a hyperabelian locally minimax group and k be a field of characteristic $p > 0$. Let $\Delta_k G = G - 1 + \mathfrak{p}$ for some right ideal $\mathfrak{p}$ of the group algebra kG , and let $G_0 = (1 + \mathfrak{p}) \cap G$. Then the maximal normal p' -subgroup of G is contained in G_0 .*

It can be shown that Proposition 3.2 extends to hyperabelian groups with finite torsion-free rank, if and only if, for every hyperabelian product $G = AB$ with finite torsion-free rank the rank equality $r_0(G) = r_0(A) + r_0(B) - r_0(A \cap B)$ holds.

Proposition 3.2 has the following consequence, which may be useful when products of two locally minimax groups are studied and which perhaps also extends to hyperabelian products of groups with finite torsion-free rank. A group G is called **locally minimax** if every finitely generated subgroup of G is a minimax group.

Corollary 3.3. *Let the group $G = AB = AM = BM$ be the product of two locally minimax subgroups A and B and a normal elementary abelian p -subgroup M with $A \cap M = B \cap M = 1$. Then the maximal normal p' -subgroup of A is normal in G .*

3.2. Hyperabelian Groups with Finite Torsion-Free Rank

In Chapter 6 of [7] it is shown that under certain finiteness conditions a triply factorized group $G = AB = AK = BK$ of two subgroups A and B and a normal subgroup K satisfies some nilpotency condition if the three subgroups A , B and K satisfy this same nilpotency condition. For instance, Amberg, Franciosi and de Giovanni have shown that a group $G = AB = AK = BK$ with finite abelian section rank is locally nilpotent (and hence hypercentral) if A , B and K are locally nilpotent ([7], Theorem 6.3.8). In the proof of this cohomological arguments play a role. The proof of the following more general result uses Theorem 3.1.

Theorem 3.4 (Amberg and Sysak [13]). *Let the hyperabelian group $G = AB = AM = BM$ be the product of three locally nilpotent subgroups A , B and M , where M is normal in G . If G has finite torsion-free rank, then G is locally nilpotent.*

If the hyperabelian group G in Theorem 3.4 does not have finite torsion-free rank, then G need not be locally nilpotent, even if A , B and M are abelian; see [7], Theorem 6.1.1. By Theorem 2.4 there exists a locally finite-soluble group $G = AB = AM = BM$ where M is a minimal normal subgroup of G and so an elementary abelian q -subgroup for some prime q and A and B are locally finite p -subgroups for some prime p , but G is not locally nilpotent. Since G is a periodic radical group, this example shows that Theorem 3.4 also becomes false when the factorized group G is no longer hyperabelian. It also follows that the Hirsch–Plotkin radical of a radical and locally finite-soluble product of two locally nilpotent subgroups need not be factorized, so that Question 11 (a) of [7] has a negative answer.

Recall that the **Hirsch–Plotkin series** of the group G is defined by

$$R_0 = 1, \quad R_{\alpha+1}/R_\alpha \text{ is the Hirsch–Plotkin radical of } G/R_\alpha$$

for every ordinal α , and for limit ordinals λ

$$R_\lambda = \bigcup_{\beta < \lambda} R_\beta.$$

Theorem 3.4 has the following consequence which is proved as in the case of a hyperabelian group with finite abelian section rank (see [7], Corollary 6.3.9).

Corollary 3.5. *Let the hyperabelian group $G = AB$ with finite torsion-free rank be the product of two locally nilpotent subgroups A and B . Then each term of the Hirsch–Plotkin series of G is factorized. In particular the Hirsch–Plotkin radical $R = R(G)$ of G is factorized, i.e. satisfies $R = (A \cap R)(B \cap R)$ and $A \cap B \subseteq R$.*

Comments on the Proof of Theorem 3.4. The proof is by induction on the torsion-free rank of M . Suppose first that the torsion-free rank of M is trivial, i.e. M is periodic.

The hypotheses of the theorem are inherited by epimorphic images. But local nilpotency is a property defined by the finitely generated subgroups of a group. To characterize this group class “from above”, the following definition turns out to be useful.

A group G is a **p -group over its subgroup S** if for every g in G there is a p -power p^n such that g^{p^n} is in S .

The relevance of this concept in our situation can be seen from the following lemma.

Lemma 3.6. *Let the hyperabelian group $G = AB = AM = BM$ be the product of two locally nilpotent subgroups A and B and a normal p -subgroup M of G . If G is a p -group over $A \cap B$, then G is locally nilpotent.*

Using this lemma it is not difficult to see that it is sufficient to prove Theorem 3.4 for a periodic normal subgroup M of G in the case when M is an elementary abelian p -group for some prime p . For this the following lemma is crucial.

Lemma 3.7. *Let G be a hyperabelian group such that $G = AM = BM = AB$, where M is an elementary abelian normal p -subgroup of G and A and B are locally nilpotent groups with finite torsion-free rank. Then G is a p -group over $A \cap B$.*

In the proof of this Theorem 3.1 and Proposition 3.2 and the following lemmas play a role.

Lemma 3.8. *Let G be a locally nilpotent group and k a field of characteristic $p > 0$ and let $\mathfrak{p}$ be a right ideal of the group algebra kG . Let $G_0 = (1 + \mathfrak{p}) \cap G$ and let G be a p -group over G_0 . Then for every finitely generated subring V of $\Delta_k(G)$ there exists a positive integer m such that $V^m \subseteq \mathfrak{p}$.*

Lemma 3.9. *Let G be a hyperabelian and locally nilpotent group with finite torsion-free rank and let k be a field of characteristic $p > 0$. Suppose that $\Delta_k(G) = G - 1 + \mathfrak{p}$ for some right ideal $\mathfrak{p}$ of the group algebra kG . Then G is a p -group over $G_0 = (1 + \mathfrak{p}) \cap G$.*

If Theorem 3.4 is false in the general case choose a counterexample where the torsion-free rank of M is minimal. Then we have seen that M is not periodic, so that the maximal periodic subgroup $T = T(M)$ is properly contained in M . It can be shown that also G/T is a counterexample, so that we may assume that M is torsion-free and hence nilpotent. We may even suppose that M is torsion-free abelian. If $T(G)$ is the maximum periodic normal subgroup of G , it follows that $M \cap T(G) = 1$. Hence $G/T(G)$ is not locally nilpotent, and we may suppose that G has no non-trivial periodic normal subgroups. In particular $C_A(M)$ is torsion-free. Since the periodic subgroups of $A/C_A(M)$ are finite, it follows that G has finite Prüfer rank, and so G is locally nilpotent by the above-mentioned theorem of Amberg, Franciosi and de Giovanni (see [7], Theorem 6.3.8). This final contradiction proves Theorem 3.4.

Theorem 3.4 raises the following questions about a hyperabelian group $G = AB = AM = BM$ with finite torsion-free rank which is the product of two subgroups A and B and a normal subgroup M of G . Note that these questions have positive answers if G has finite abelian section rank (see [5], [6] and [20]):

(a) If A , B and M are (locally nilpotent)-by- $\mathfrak{X}$, is it true that also G (locally nilpotent)-by- $\mathfrak{X}$, where $\mathfrak{X}$ is the class of finite, Chernikov, polycyclic, minimax, periodic groups or the class of groups with finite Prüfer rank or finite abelian section rank?

(b) If A , B and M are locally polycyclic, is it true that also G locally polycyclic?

(c) If A , B and M are FC -hypercentral (FC -nilpotent), is it true that G likewise FC -hypercentral (FC -nilpotent)?

3.3. FC-Nilpotent Groups

Another result about triply factorized groups is the following theorem concerning *FC*-nilpotent groups. Recall that a group G is ***FC-nilpotent*** if the upper *FC*-central series of G reaches G after a finite number of steps.

Theorem 3.10 (Amberg, Franciosi and de Giovanni [8]). *Let the FC-nilpotent group $G = AB = AM = BM$ be the product of two hypercentral subgroups A and B and a nilpotent normal subgroup M of G . Then G is hypercentral.*

It is well-known that for *FC*-nilpotent groups the classes of hypercentral and of locally nilpotent groups coincide. It seems likely that every *FC*-hypercentral group $G = AB = AM = BM$ which is the product of three hypercentral subgroups A , B and M , where M is normal in G , is likewise hypercentral.

Theorem 3.10 has the following consequence on the “outside problem” (see section 7 of [7]).

Corollary 3.11. *Let the FC-nilpotent group $G = AB$ be the product of a nilpotent proper subgroup A and a proper hypercentral subgroup B . Then either A or B is contained in a proper normal subgroup of G . If A and B are distinct hypercentral subgroups, then either A or B is contained in a proper ascendant subgroup of G .*

3.4. Connections with Radical Rings

An associative ring R is called ***radical*** if it coincides with its Jacobson radical, which means that R forms a group under the operation $a \circ b = a + b + ab$ for all a and b in R . This group is called the ***adjoint group*** R° of R . If the radical ring R is embedded in the usual way into a ring R_1 with identity, then R° is isomorphic with the subgroup $1 + R$ of the group of units of R_1 . Examples of such radical rings are the nilpotent rings and even the nil rings. Recall that an associative ring R is a ***nil ring*** if every element $a \in R$ is ***nilpotent***, i.e. there exists a positive integer n such that $a^n = 0$. The ring R is ***nilpotent*** if $R^n = 0$ for some positive integer n .

The adjoint group $A = R^\circ$ operates on the additive group $M = R^+$ of R via the rule $m^a = m + ma$ for all a in A and m in M . Let $G(R) = A \ltimes M$ be the semidirect product of A with M and identify A and M with its subgroups $\{(a, 0) \mid a \in A\}$ and $\{(0, m) \mid m \in M\}$, respectively. If $B = \{(r, r) \mid r \in R\}$ is the diagonal subgroup of $G(R)$, then the ***associated group*** of R is

$$G(R) = A \ltimes M = B \ltimes M = AB.$$

Then $G = AB$ and $A \cap B = 1$, where the subgroups A and B are isomorphic with R° and the normal subgroup M of $G(R)$ is isomorphic with R^+ . This construction gives examples of triply factorized groups with the intersection property.

What can be said about the structure of the adjoint group of a radical ring? The following theorem, which generalizes a result of Watters in [56] for the maximum condition, gives a relation between the three groups R° , R^+ and $G(R)$ of a radical ring.

Theorem 3.12 (Amberg and Dickenschied [4]). *Let $\mathfrak{X}$ be a class of minimax groups which is closed under the forming of subgroups, epimorphic images and extensions. The following conditions of the radical ring R are equivalent:*

- (i) *The additive group R^+ is an $\mathfrak{X}$ -group.*
- (ii) *The associated group $G(R)$ is an $\mathfrak{X}$ -group.*
- (iii) *The adjoint group R° is an $\mathfrak{X}$ -group.*

In this case R is a nilpotent ring.

The additive group of the (commutative) radical subring

$$R = \left\{ \frac{u}{v} \mid u \text{ is an even and } v \text{ is an odd integer} \right\}$$

of the ring of rational numbers has Prüfer rank 1, but R° has infinite torsion-free rank and R is not even a nil ring. Hence there is no complete analogue of Theorem 3.12 for the other finiteness conditions. However, the following theorem shows that these finiteness conditions are inherited from the adjoint group R° of a radical ring R to the additive group R^+ and that they imply some nilpotency conditions of the ring R .

Theorem 3.13 (Amberg and Dickenschied [4]). *Let R be a radical ring. Then the following holds:*

- (a) *If R° has finite torsion-free rank n , then also $r_0(R^+) = n$ and R is a nil ring.*
- (b) *If R° has finite abelian subgroup rank, then so does R^+ , and R is a two-sided T -nilpotent ring of class $cl(R) \leq \omega + r_0(R^+)$.*
- (c) *If R° has finite Prüfer rank, then so does R^+ , and $r(R^+)$ is bounded by a function which only depends on $r(R^\circ)$.*

Here a ring R is called **two-sided T -nilpotent of class $cl(R) = \alpha$** if $B_\alpha(R) = R$ and α is the least ordinal with this property, where the **transfinite two-sided annihilator series** of a ring R is defined by

$$B_0(R) = 0, \quad B_{\alpha+1}(R) = \{a \in R \mid aR + Ra \subseteq B_\alpha(R)\}$$

for each ordinal α and for each limit ordinal λ

$$B_\lambda(R) = \bigcup_{\beta < \lambda} B_\beta(R).$$

For nil-rings and for radical rings whose additive group is periodic more can be said, in particular the converse of Theorem 3.13 holds in these cases (see [19]). The question arises which is the exact bound for the Prüfer rank in Theorem 3.13 (c). Is perhaps $r(R^+) \leq 2r(R^\circ)$?

3.5. On Products of Abelian Groups

If the group $G = AB = AM = BM$ is the product of three abelian subgroups A , B and K where K is normal in G such $A \cap M = B \cap M = 1$, then there exists a radical ring R such that the group G is the associated group $G(R)$ which can be constructed as the semidirect product of the additive group $R^+ \simeq K$ of R by the adjoint group $R^\circ \simeq A$ of R (see [7], Proposition 6.1.4). Using this the following theorem has been proved which answers [2], Question IX (a) in the affirmative.

Theorem 3.14 (Sysak [52]). *Let the group $G = AB$ be the product of two abelian subgroups A and B .*

- (a) *The Fitting subgroup of G is factorized if and only if the Hirsch–Plotkin radical of G is factorized.*
- (b) *If A or B has finite torsion-free rank, then the Fitting subgroup and the Hirsch–Plotkin radical of G are factorized.*

4. Products of Nilpotent Groups

4.1. Finite Products of Nilpotent Groups

A well-known theorem of Kegel and Wielandt states that a finite group $G = AB$ which is the product of two nilpotent subgroups A and B is soluble (see [7], Theorem 2.4.3). In order to determine the structure of such groups it is interesting to know which subgroups of G are conjugate to a subgroup S that inherits the factorization. This question is discussed in detail in [10]. A subgroup S of a factorized group $G = AB$ is called **prefactorized** if $S = (A \cap S)(B \cap S)$; it is called **factorized** if in addition $A \cap B$ is contained in S . For instance it follows from Corollary 3.5 that the Fitting subgroup of a finite product of two nilpotent groups is always factorized. On the other hand, it can be shown that unlike the Fitting subgroup the commutator subgroup, the Frattini subgroup, the centre and the hypercentre of a finite product of nilpotent groups are in general not prefactorized.

The following elementary lemma is easily proved by using the theory of P.Hall on finite soluble groups.

Lemma 4.1. *Let the finite and soluble group $G = AB$ be the product of two subgroups A and B whose orders are relatively prime. Then every subgroup of G has a factorized conjugate. In particular every subnormal subgroup of G is factorized.*

Recall that a set Σ of Hall subgroups of a finite soluble group G is a **Hall system** of G if Σ contains a Hall π -subgroup of G for every set of primes π and $HK = KH$ for every pair H and K of subgroups in Σ .

Lemma 4.2. *If the finite soluble group $G = AB$ is the product of two nilpotent subgroups A and B , then the system Σ of all $A_\pi B_\pi$ where π is a set of primes, is a Hall system of G which reduces into every prefactorized subgroup S of G , i.e. the set of subgroups $S \cap A_\pi B_\pi$ is a Hall system of S .*

Using this the following uniqueness statement about pronormal subgroups can be proved. Recall that a subgroup P of a group G is **pronormal** if P and P^g are conjugate in $\langle P, P^g \rangle$ for every element g in G .

Proposition 4.3. *A pronormal subgroup of a finite group $G = AB$ which is the product of two nilpotent subgroups A and B has at most one prefactorized conjugate.*

For abnormal subgroups we even obtain the following existence theorem. A subgroup S of a group G is **abnormal** if $g \in \langle S, S^g \rangle$ for every element g in G .

Theorem 4.4 (Amberg and Höfling [10]). *Every abnormal subgroup of a finite group $G = AB$ which is the product of two nilpotent subgroups A and B has a unique factorized conjugate.*

The proof of Theorem 4.4 is by induction on the order of the group G . It is shown that a minimal counterexample is a nonabelian primitive group and so the orders of A and B are relatively prime by [7], Lemma 2.5.2.

A class $\mathfrak{H}$ of finite soluble groups is a **Schunck class** if a group G is an $\mathfrak{H}$ -group whenever its primitive epimorphic images are $\mathfrak{H}$ -groups. For instance every saturated formation is a Schunck class. The characteristic $\text{Char}(\mathfrak{H})$ is the set of all primes p such that the cyclic group of order p is an $\mathfrak{H}$ -group. Furthermore, a subgroup Y of a group G is $\mathfrak{H}$ -**maximal** if for every $\mathfrak{H}$ -subgroup Y^* of G with $Y \subseteq Y^*$ it follows that $Y = Y^*$.

Theorem 4.5 (Amberg and Höfling [10]). *Let $\mathfrak{H}$ be a Schunck class and let the finite group $G = AB$ be the product of two nilpotent subgroups A and B . If H is an $\mathfrak{H}$ -maximal subgroup of G , then*

- (a) *if $\pi(A) \cap \pi(B) \subseteq \text{Char}(\mathfrak{H})$, then H has a factorized conjugate in G ,*
- (b) *if $\mathfrak{H}$ is a saturated formation, then H has a prefactorized conjugate.*

As a consequence of Proposition 4.3 and Theorem 4.5 we obtain the following generalization of a result of Heineken; see [7], Theorem 2.5.15. Recall that a subgroup P of a finite group G is an $\mathfrak{H}$ -**projector** of G if the factor group PN/N is $\mathfrak{H}$ -maximal in G/N for every normal subgroup N of G .

Corollary 4.6. *Let $\mathfrak{H}$ be a Schunck class and let the finite group $G = AB$ be the product of two nilpotent subgroups A and B .*

- (a) *If $\pi(A) \cap \pi(B) \subseteq \text{Char}(\mathfrak{H})$, then G has a unique factorized $\mathfrak{H}$ -projector.*
- (b) (Heineken [26]) *If $\mathfrak{H}$ is a saturated formation, then G has a unique prefactorized $\mathfrak{H}$ -projector.*

The following corollary of Theorem 4.5 extends earlier results of Kegel, Pennington and Peterson; see [7], Theorem 2.5.10.

Corollary 4.7 (Amberg and Fransman [9]). *Let $\mathfrak{H}$ be a Schunck class of finite groups. Let the finite group $G = AB = AC = BC$ be the product of two nilpotent subgroups A and B and an $\mathfrak{H}$ -subgroup C . If $\pi(A) \cap \pi(B) \subseteq \text{Char}(\mathfrak{H})$, then G is a soluble $\mathfrak{H}$ -group.*

In [10] an example is given that for Fitting classes $\mathfrak{F}$ the $\mathfrak{F}$ -injectors of a finite product of nilpotent groups need not be prefactorized in general. However it is also shown there that for Fitting classes $\mathfrak{F}$ every product of two finite nilpotent groups has a unique prefactorized (factorized) $\mathfrak{F}$ -injector if and only if every product of two finite nilpotent groups has a prefactorized (factorized) $\mathfrak{F}$ -radical.

Some results of this section have been extended in the forthcoming dissertation of B. Höfling to certain locally finite-soluble products of two locally nilpotent subgroups.

4.2. On the Derived Length

It is a well-known open question whether the derived length of a finite group $G = AB$ which is the product of two nilpotent subgroups A and B is bounded by a function of the nilpotency classes c of A and d of B (see [7], Question 5). Gross proved that the Frattini factor group $G/\Phi(G)$ has derived length at most $c + d$, and this of course means that the subgroup $G^{(c+d-1)}$ is nilpotent; see [7], Theorem 2.5.5. The following theorem extends this result to radical groups with finite abelian section rank. By $\mathfrak{S}_1$ we denote the class of radical groups with finite abelian section rank containing elements of only finitely many distinct prime orders.

Theorem 4.8 (Franciosi and de Giovanni [23]). *Let the radical group $G = AB$ with finite abelian section rank be the product of two nilpotent subgroups A and B with nilpotency classes c and d , respectively. Then the subgroup $G^{(c+d-1)}$ is locally nilpotent. If G is an $\mathfrak{S}_1$ -group, then $G^{(c+d-1)}$ is nilpotent.*

Pennington has shown that if the finite group $G = AB$ is the product of two nilpotent subgroups A and B with nilpotency classes c and d , then the $(c + d)$ -th term of the derived series of G is a nilpotent π -group where $\pi = \pi(A) \cap \pi(B)$; see [7], Theorem 2.5.3. This result can be extended to periodic radical products of two nilpotent groups.

Theorem 4.9 (Franciosi and de Giovanni [23]). *Let the periodic radical group $G = AB$ be the product of two nilpotent subgroups A and B with nilpotency classes c and d , respectively. Then the subgroup $G^{(c+d)}$ is a π -group, where $\pi = \pi(A) \cap \pi(B)$. In particular, if $\pi(A) \cap \pi(B) = \emptyset$, the group G is soluble with derived length at most $c + d$.*

More on the derived length of finite soluble factorized groups can be found in Kazarin [34].

4.3. Hirsch–Plotkin Radical and Fitting Subgroup

It follows for instance from Corollary 3.5 that the Fitting subgroup of a finite product of two nilpotent groups is factorized. In this section we mention some related results on more general classes of soluble products of nilpotent groups.

The first theorem shows that if $G = AB$ is a radical group with finite abelian section rank which is factorized by two nilpotent subgroups A and B , then the centres $Z(A)$ and $Z(B)$ act trivially on the second factor of the Hirsch–Plotkin series of G .

Theorem 4.10 (Franciosi and de Giovanni [23]). *Let the radical group $G = AB$ with finite abelian section rank be the product of two nilpotent subgroups A and B . If R is the Hirsch–Plotkin radical of G and H/R is the Hirsch–Plotkin radical of G/R , then $[Z(A), H] \subseteq R$.*

Theorem 4.10 has the following consequence.

Corollary 4.11. *Let the radical group $G = AB$ with finite abelian section rank be the product of two nilpotent subgroups A and B , and let R be the Hirsch–Plotkin radical of G . Then the subgroups $Z(A)R$ and $Z(B)R$ are subnormal in G with defect at most 2.*

However, it can be shown that there exists a finite group $G = AB$ which is the product of two nilpotent subgroups A and B with Fitting subgroup $F = F(G)$ such that $Z(A)F$ and $Z(B)F$ are both not normal in G , and the subgroups AF and BF are both not subnormal in G . Corresponding to Theorem 4.10 we have the following result.

Theorem 4.12 (Franciosi and de Giovanni [23]). *Let the $\mathfrak{S}_1$ -group $G = AB$ be the product of two nilpotent subgroups A and B . If F is the Fitting subgroup of G and L/F is the Fitting subgroup of G/F , then $[Z(A), L] \subseteq F$.*

It can be deduced from Theorem 4.12 that if the $\mathfrak{S}_1$ -group $G = AB$ is the product of two nilpotent subgroups A and B , then the subgroups $Z(A)F$ and $Z(B)F$ are subnormal of defect at most 2. Moreover, $[Z(A), Z(B)]$ is a nilpotent subnormal subgroup of G .

5. Locally Finite Products of Locally Nilpotent Groups

5.1. Minimal Normal Subgroups

Robinson and Stonehewer have shown in [43] that every chief factor of a product $G = AB$ of two abelian subgroups A and B is centralized by A or B ; see [7], Theorem 6.6.9. Moreover, Stonehewer proved in [45] that if $G = AB$ is finite and A and B are nilpotent, then for every minimal normal subgroup N of G one of the subgroups AN and BN is nilpotent. The following theorem extends these results to periodic radical groups factorized by two locally nilpotent subgroups.

Theorem 5.1 (Franciosi, de Giovanni and Sysak [24]). *Let the periodic radical group $G = AB$ be the product of two locally nilpotent subgroups A and B , at least one of which is hyperabelian. If N is a minimal normal subgroup of G , then one of the subgroups AN and BN is locally nilpotent. Moreover, if AN is locally nilpotent, then $Z(A)$ centralizes N , and if both AN and BN are locally nilpotent the subgroup N is contained in the centre of G .*

Theorem 5.1 has the following consequence for trifactorized groups.

Corollary 5.2. *Let the periodic radical group $G = AB = AC = BC$ be the product of three locally nilpotent subgroups A , B and C , at least two of which are hyperabelian. Then G is locally nilpotent.*

It follows from Theorem 2.4 that in these results the hypothesis that the factors are hyperabelian cannot be omitted.

The next result shows that Theorem 4.10 also holds for periodic hyperabelian groups.

Theorem 5.3 (Franciosi, de Giovanni and Sysak [24]). *Let the periodic radical group $G = AB$ be the product of two locally nilpotent subgroups A and B , at least one of which is hyperabelian. If R is the Hirsch–Plotkin radical of G and H/R is the Hirsch–Plotkin radical of G/R , then $[Z(A), H] \subseteq R$.*

It follows from Theorem 5.3 that in this situation the subgroups $Z(A)R$ and $Z(B)R$ are subnormal with defect at most 2. Moreover, the subgroup $[Z(A), Z(B)]$ is contained in the Hirsch–Plotkin radical of G and so is locally nilpotent.

It is also possible to extend the theorem of Gross to periodic radical groups.

Theorem 5.4 (Franciosi, de Giovanni and Sysak [24]). *Let the periodic radical group $G = AB$ be the product of two nilpotent subgroups A and B with nilpotency classes c and d , respectively. Then the subgroup $G^{(c+d-1)}$ is locally nilpotent. In particular, the Hirsch–Plotkin series of G has length at most $c + d$.*

5.2. An Extension of the Theorem of Kegel and Wielandt

In which ways does the Theorem of Kegel and Wielandt extend to infinite groups, in particular to locally finite groups? Is for example every locally finite product of two locally nilpotent subgroups always locally soluble? Kegel has shown in [36] that every linear product of two nilpotent groups is soluble and that every locally finite product of two locally nilpotent *FC*-groups is locally soluble. Moreover, Kazarin proved in [33] that if the locally finite group $G = AB$ is the product of an abelian group A and a locally nilpotent group B , then G is locally soluble.

Theorem 5.5 (Franciosi, de Giovanni and Sysak [25]). *Let the locally finite group $G = AB$ be the product of two hypercentral subgroups A and B , at least one of which is an *FC*-group. If R is the Hirsch–Plotkin radical of G , then G/R is hyperabelian. In particular, G is a radical group.*

Theorem 5.5 implies another generalization of Pennington's theorem on finite products of nilpotent groups of coprime order (see [7], Theorem 2.5.3, and Theorem 4.9 above).

Corollary 5.6. *Let the locally finite groups $G = AB$ be the product of two nilpotent subgroups A and B , with classes c and d , respectively. If one of the two subgroups A or B is an *FC*-group, then the subgroup $G^{(c+d)}$ is a π -group, where $\pi = \pi(A) \cap \pi(B)$. In particular, if $\pi(A) \cap \pi(B) = \emptyset$, the group G is soluble with derived length at most $c + d$.*

6. Some Further Results

6.1. On the Rank of a Product of Two Finite p -Groups

It is known that the Prüfer rank of a finite p -group $G = AB$ which is the product of two subgroups A and B whose Prüfer ranks are bounded by r , is bounded by a polynomial function of r ; see [7], Theorem 4.3.5. Although no bound is given there explicitly, the analysis of the proof of this theorem leads to polynomial bounds of relatively high degree. It seems to be unknown whether there is a linear bound. An attempt to give better bounds is made in [11].

The following theorem gives a bound for the normal rank of $G = AB$ which will immediately give a bound for the Prüfer rank of G essentially by a theorem of Thompson, [29], Satz 12.3, p. 343. Recall that the **normal rank** $r_n(X)$ of a group X is the maximum of the minimal number of generators of each normal subgroup of X . If α is a real number, then $[\alpha] = m$ is the smallest integer such that $\alpha \leq m$. δ_{ij} denotes the Kronecker symbol.

Theorem 6.1 (Amberg and Kazarin [11]). *Let the finite p -group $G = AB$ be the product of two subgroups A and B . Let $r^* = \min\{r(A), r(B)\}$ and $r_1 = r(A) + r(B)$. Then the normal rank $r_n(G)$ satisfies the following inequality:*

$$r_n(G) \leq r^* (\lceil \log_p r_n(G) \rceil + 1 + \lceil \log_p r^* \rceil \lceil \log_2 2r^* \rceil + \delta_{2p}) + r_1.$$

The inequality in Theorem 6.1 implies that for any $\varepsilon > 0$ and for sufficiently large r^* we have the following almost linear bound

$$r_n(G)^{1-\varepsilon} \leq r^* (3 + \lceil \log_p r^* \rceil \lceil \log_2 2r^* \rceil) + r_1.$$

Therefore the bound in Theorem 6.1 for the normal rank of G and so also the bound for its Prüfer rank is close to being linear. Even if this result may not be best possible, it will be useful in the study of the structure of finite products of groups with low rank.

If the two subgroups A and B are abelian, Theorem 6.1 can be improved as follows.

Theorem 6.2 (Amberg and Kazarin [11]). *Let the finite p -group $G = AB$ be the product of two abelian subgroups A and B . Let $r^* = \min\{r(A), r(B)\}$ and $r_1 = r(A) + r(B)$. Then the normal rank $r_n(G)$ of G satisfies the inequality*

$$r_n(G) \leq r^* \lceil \log_p r_n(G) \rceil + r_1.$$

If the finite p -group $G = AB$ is the product of two cyclic subgroups A and B , then it follows from Theorem 6.2 that $r_n = r_n(G) \leq \lceil \log_p r_n \rceil + 2$. This implies $r_n \leq 3$ and even $r_n \leq 2$ for $p > 3$. Note however that there exists a finite 2-group of normal rank 3 which is a product of two of its cyclic subgroups. (see [29], Aufgabe 28, p. 341).

6.2. More on the Hirsch–Plotkin Radical

The symmetric group of degree 4 is the product of a subgroup isomorphic with the symmetric group of degree 3 and a cyclic group of order 4, but its Fitting group $F = F(G)$ is not factorized, i.e. $F \neq AF \cap BF$. But even in this general situation something more can be said. Recall that by [7], Lemma 2.1.2, for the FC-centre of an arbitrary product $G = AB$ of two subgroups A and B the following holds: If A_1 , B_1 and F_1 are the FC-centres of A , B and G , then $F_1 = A_1 F_1 \cap B_1 F_1$. Similar statements hold for the centre and the hypercentre.

The following theorem is first proved for the special case of a finite soluble product of two subgroups.

Theorem 6.3 (Franciosi and de Giovanni [21]). *Let the soluble-by-finite group $G = AB$ with finite abelian section rank be the product of two subgroups A and B , and let R be the Hirsch–Plotkin radical of G . Then $R = A_0 R \cap B_0 R$, where A_0 and B_0 are the Hirsch–Plotkin radicals of A and B , respectively.*

If the two subgroups A and B in Theorem 6.3 are locally nilpotent, then it follows in particular that the Hirsch–Plotkin radical of G is factorized. Similarly, the following result implies that the Fitting subgroup of a soluble $\mathfrak{S}_1$ -group which is the product of two nilpotent subgroups, is factorized.

Theorem 6.4 (Franciosi and de Giovanni [21]). *Let the soluble-by-finite $\mathfrak{S}_1$ -group $G = AB$ be the product of two subgroups A and B , and let F be the Fitting subgroup of G . Then $F = A_0F \cap B_0F$, where A_0 and B_0 are the Fitting subgroups of A and B , respectively.*

By [7], Theorem 6.1.2, there exists a triply factorized group $G = AB = AK = BK$, where A , B and K are torsion-free abelian subgroups and K is normal in G , but G is not locally nilpotent. Hence Theorem 6.3 does not hold in general. But the question arises whether the above theorems can be extended to soluble-by-finite groups with finite torsion-free rank, in particular to periodic groups.

A result of Johnson [32] says that for every arbitrary finite soluble group $G = AB$ which is the product of two subgroups A and B and for every set of primes π the maximal π -subgroups satisfy $O_\pi(A) \cap O_\pi(B) \subseteq O_\pi(G)$. While this is false for finite groups in general it even holds for arbitrary periodic hyperabelian groups with no perfect subgroups and no infinite elementary abelian groups involved in $O_\pi(A)$.

In [46] Sozutov has shown that there exist groups $G = AB$ which are the product of two normal subgroups A and B in which every subgroup generated by $n \geq 2$ elements is nilpotent, but G does not have this property. Thus, in particular, the product of two normal 2-nilpotent groups need not be 2-nilpotent. Observe however that by the theorem of Hirsch and Plotkin the product of two normal locally nilpotent subgroups is always locally nilpotent.

6.3. Groups Factorized by Finitely Many Subgroups

By the theorem of Kegel and Wielandt every finite group $G = A_1A_2 \dots A_n$ which is the product of pairwise permutable nilpotent subgroups A_i is soluble (see [7], Corollary 2.4.4). It was proved in [36] that a linear group $G = A_1A_2 \dots A_n$ which is the product of pairwise permutable locally nilpotent subgroups A_i is soluble in each of the following cases: (1) G is finitely generated, (2) $n = 2$, (3) $n \geq 3$ and the product of any three of the subgroups A_i is soluble. Kegel asked in [36], Remark to Theorem 2.8, whether there are insoluble linear groups over a field which are the product of finitely many pairwise permutable locally nilpotent subgroups. Such groups can in fact be constructed in the following way.

Let R be an associative ring and R_1 the ring obtained from R by adjoining an identity 1 if R is without one and $R_1 = R$ otherwise. Denote by $G_2(R)$ the group of all invertible 2×2 -matrices over R_1 that are congruent to the identity matrix modulo R , and by $t_{ij}(s)$ the transvection with the element s in the position (i, j) . Then the